

REVIEW

Exploring the application of ferroelectret nanogenerators in medical engineering

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Abstract

Flexible devices derived from piezoelectric materials have gained considerable attention due to their exceptional biocompatibility. Among these, ferroelectret nanogenerators (FENG) is a novel type of flexible piezoelectric device that integrates self-powering, actuation, and sensing capabilities. Its potential applications within the realm of human physiological medicine are continuously expanding. Given the increasing public emphasis on health, demand is escalating for devices that can monitor human activity indicators and reflect the state of bodily functions. In this context, flexible devices with superior biocompatibility have become increasingly valuable for medical testing, sports training guidance, and physical exercise. Due to its exceptional flexibility, high sensitivity, significant piezoelectric coefficient, and excellent biocompatibility, FENG has garnered considerable attention in medical applications, demonstrating strong potential for accurately detecting human physiological activities. Therefore, an in-depth exploration of FENG's applications in medical testing and auxiliary treatment carries significant practical implications. However, current research on FENG's application in the medical field lacks comprehensive understanding and systematic evaluation. This paper aims to review the most recent advancement in the application of FENG in medical settings. By presenting a wide variety of application examples and systematic evaluations, we aim to demonstrate that FENG can meet the personalized needs of the medical application field. First, this paper introduces the working principle of FENG and its fabrication methods, followed by an introduction to FENG's applications in four major human physiological systems: blood circulation, respiration, muscle movement, and nerve reflexes. Finally, potential directions for further development of FENG and the challenges faced are discussed.

KEYWORDS

blood circulation, ferroelectret nanogenerators, medical applications, muscle activity, neural reflex, physiological movement monitoring, respiratory system

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1 | INTRODUCTION

Physiological movements encompass a range of essential life-sustaining activities, including respiratory circulation, blood circulation, and nerve reflexes. More specifically, physiological movement refers to the dynamic changes that occur within the body's internal organ systems in adherence to certain physiological laws and rhythms during various activities. This process necessitates the collaborative involvement and precise regulation of several crucial systems, such as the cardiovascular,¹ respiratory,² nervous,³ and muscular systems,⁴ to collectively fulfill different motor functions and body goals. Physiological movement is not merely a fundamental form of movement essential for human survival, but also a vital aspect of maintaining body homeostasis and normal physiological functions. Consequently, the accurate observation of physiological movement and the implementation of effective auxiliary improvement measures have demonstrated significant potential and value in medical treatment^{5,6} and healthcare^{7,8} fields.

Physiological movement monitoring's⁹ core rests on a comprehensive understanding and evaluation of physiological changes occurring during human movement. These changes encompass several aspects, including energy metabolism,¹⁰ cardiovascular system response,¹¹ thermal regulation mechanisms,¹² and respiratory system dynamics.¹³ This information is vital for directing training strategies, health management plans, and preventive measures against diseases. It helps to elucidate physiological mechanisms during exercise and effectively assess training effects and individual functional status through methods such as heart rate (HR) monitoring,¹⁴ blood pressure measurement,¹⁵ and blood biochemical index analysis.¹⁶

As living standards improve and global aging intensifies, public health awareness has grown, driving the potential of physiological exercise monitoring in areas like chronic disease prevention, acute disease alerting, population health assessment, and personalized medical treatment. Although traditional measurement methods such as electrocardiogram (ECG),¹⁷ electromyogram (EMG),¹⁸ and electroencephalogram (EEG),¹⁹ utilize flexible non-invasive electrodes and play a significant role in flexible electronics, these electrodes, when in contact with human skin, can effectively track surface electromyography signals. These signals are used to analyze human activities, such as facial expressions and body movements, and to control external devices. While these devices are convenient to wear, they hinder the physiological functions of the skin, such as sweating and heat

dissipation, which limits their application in daily health monitoring. In this context, the creation of flexible, lightweight, and user-friendly health monitoring devices has taken center stage in current research. Recent years have seen the rise of novel sensing technologies, including capacitive flexible pressure sensors,²⁰ piezoresistive flexible pressure sensors,²¹ strain sensors,²² and generalized flexible sensors.²³ Particularly, capacitive and piezoresistive flexible pressure/strain sensors,²⁴ with their lightweight and high stretchability features, have opened up unprecedented opportunities in the electro-mechanical biosensing field.²⁵ These sensors can precisely capture various physiological signals, including joint movements,²⁶ speech attributes,²⁷ breathing patterns,²⁸ and heartbeats,²⁹ offering robust support for comprehensive health monitoring.

To further enhance monitoring efficiency and user experience, body area sensor networks (BASNs)^{30,31} have emerged. These networks integrate a variety of flexible sensors distributed across different body parts, aiming to construct an efficient and integrated health information collection system. However, most flexible sensors currently rely on external power sources, which not only increases the user's carrying burden, but also exacerbates energy consumption issues.³² Moreover, the manufacturing of capacitive and resistive pressure/strain sensors typically involves complex laser ablation processes to create micro-nano structures on the material surface, leading to high manufacturing costs and process complexity. Therefore, exploring more efficient, economical, and environmentally friendly sensor fabrication techniques, as well as developing self-powered or low-power consumption flexible sensor systems, will be an important direction for future research.

Since the introduction of the nanogenerators concept in 2006, a new trajectory for the development of flexible sensors has been set.³³ Nanogenerators are primarily classified into three categories based on their functional mechanism: thermoelectric nanogenerators,³⁴ piezoelectric nanogenerators (PENG),³⁵ and electrostatic nanogenerators, which include triboelectric nanogenerators (TENG).^{36,37} PENG can further be divided into ferroelectric nanogenerators (FENG)³⁸ and piezoelectric ceramic nanogenerators.³⁹ Among these, thermoelectric nanogenerators are comparatively more expensive. While TENG has been utilized for human movement recognition tasks such as facial movement,⁴⁰ speech,⁴¹ heart,⁴² and respiratory observation,²⁵ it has shown promising growth in the potential field of biomechanical energy harvesting.^{16,43–45} However, the inherent capacitance of TENG results in a large internal impedance (several

megaohms) and a low short current output (μA level), which limits energy transfer efficiency. The requirement for specific materials and intricate structural design to efficiently convert the limited ultrasonic energy in biological tissues further constrains its application.^{46–48} For instance, the strategy of coupling TENG with mxene hydrogel-based generators has been suggested to enhance ultrasonic energy collection.⁴⁹ Despite its heightened sensitivity to ultrasonic response and excellent biocompatibility, there is still a need for improvement in energy conversion efficiency.

Compared to TENG, FENG stands out as a unique innovation that combines self-powering, actuator, and sensor functionalities. This novel nanogenerator has found widespread applications in various domains such as keyboards,⁵⁰ microphones,^{50,51} speakers,⁵² and fitness trackers⁵³ due to its simple manufacturing process, easy stretchability, and high precision. Its potential in medical applications is also noteworthy. However, despite the abundance of reviews on nanogenerators, there is a dearth of comprehensive reviews specifically focusing on the application of FENG in the medical field. This paper aims to fill this void by reviewing the fundamental principles of FENG and its growth in medical applications.

This paper initially provides an overview of the preparation methods and operating principles of FENG. It then delves into an in-depth discussion on the varied applications of FENG in the medical field, including but not limited to blood circulation monitoring, respiratory cycle analysis, muscle movement assessment, and neuroreflex testing (Figure 1). Finally, this paper addresses the current challenges and potential future research directions concerning FENG applications, thereby offering a comprehensive research guide for researchers interested in FENG. Moreover, it provides insights and references that could shape the future development of FENG applications in medical applications.

2 | BASIC PRINCIPLES OF FENG IN MEDICAL APPLICATIONS

The accurate monitoring of physiological movements serves as an effective means of assessing an individual's physical condition, necessitating the application of high-precision sensors that are particularly suitable for human movement monitoring. In this context, FENG, an advanced sensor, has demonstrated exceptional potential. Ferroelectrets, also referred to as piezoelectrets, exhibit pronounced piezoelectric effects, a characteristic that

stems from their unique physical properties. This material, a thin film of polymer, features micro-void structures that house directionally arranged “giant dipoles”. Piezoelectrets can be divided into two categories according to their polarization attributes: unipolar and bipolar piezoelectrets. Specifically, unipolar piezoelectrets have polarized charges on only one surface of their microvoids, while the other surface lacks a polarized charge distribution.⁵⁴ In contrast, bipolar piezoelectrets' microvoid walls bear charges of opposing polarity on both surfaces. This characteristic enhances their performance in energy conversion and sensor applications, placing them at an advantage.^{55–58}

Ferroelectret materials not only exhibit a significant piezoelectric effect that rivals ceramic materials, but they also encompass unique characteristics inherent to polymer materials. These characteristics include low acoustic impedance, superior flexibility, low density, the capability for large-area film formation, and cost-effective manufacturing. Crucially, ferroelectrets, with their exceptional ability to convert between electrical and mechanical energy and their relatively low Young's modulus, display impressive performance in energy harvesting applications. Specifically, their low Young's modulus imparts FENGs with a high level of softness, predisposing the material to deformation under stress, meaning it can be more easily bent or stretched. This attribute enables FENGs to be integrated into human skin for energy storage purposes.

Moreover, ferroelectrets exhibit a powerful inverse piezoelectric effect. When the charge or voltage state on the ferroelectret material's surface alters, it propels the colossal dipoles within the FENG to relocate, thus triggering a significant change in the FENG's thickness. Consequently, FENGs have demonstrated immense application potential across various fields, including energy harvesting, sensor manufacturing, and actuator design.

2.1 | Working principle

As an emerging piezoelectric device, FENG has demonstrated significant efficacy in the mutual conversion process between electrical and mechanical energy.^{59,60} The piezoelectric effect specifically denotes the physical phenomenon where the internal charge distribution of certain specific materials alters when subjected to mechanical stress. Conversely, changes in the charge distribution or potential state on the material's surface can prompt adjustments in the material's shape. This unique

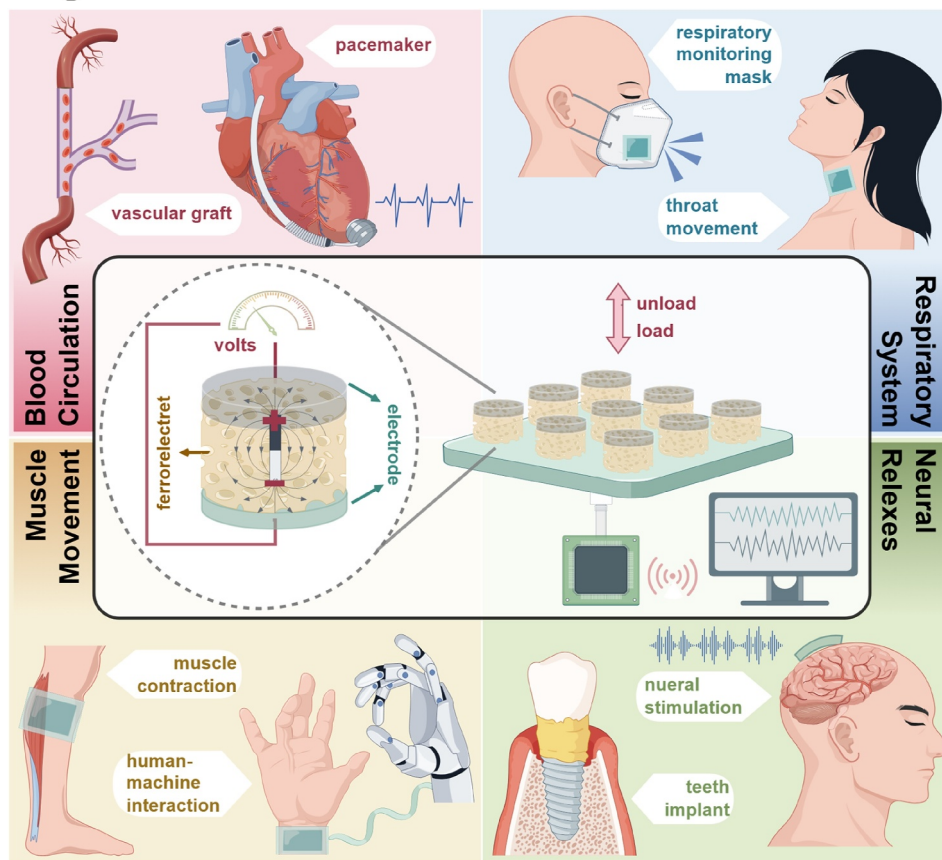


FIGURE 1 Categories defined for the medicine applications of ferroelectret nanogenerators.

attribute positions piezoelectric materials as an ideal candidate for the construction of ferroelectric nanogenerators.

Polypropylene (PP) is taken as an example. As a prospective material for FENG production,⁶¹ it is preferred not only for its superior thermal insulation properties, but also for its capacity to prepare foam structures in the PP film via micro-plasma discharge technology, thus accomplishing special functions. In this procedure, opposite charges are induced and accumulated on the upper and lower surfaces of artificially manufactured pores, resulting in numerous highly oriented large dipoles. Subsequently, when two layers of metal are sputtered on the two external surfaces of the PP film, these large dipoles induce charges of opposite polarity in each metal film.

When FENG is subjected to compression or expansion deformation along its thickness direction, its internal dipole moment dynamically responds, adjusting according to the applied pressure. Specifically, the dipole moment p is given by $p = ql$, where q denotes the positive charge borne by the hole post-polarization, and l signifies the linear distance between the positive and negative

charge centers. Consequently, when FENG experiences compression along the thickness or stretching along the length, the distance l between the positive and negative charge centers decreases correspondingly, resulting in a reduction of the dipole moment p . This decrease directly undermines FENG's electron-binding capability. This dynamic modulation of the dipole moment facilitates the compensating electron migration between the negative and positive charge electrodes. Under open circuit conditions, this migration process results in a potential difference between the two electrodes; under closed circuit conditions, it engenders charge flow or current. Conversely, when the surface electrode's charge density alters, this modification can reciprocally reconfigure FENG's substantial dipole structure, exhibiting a type of inverse electromechanical interaction effect. The cross-section of the PP film reveals the alternating arrangement of the solid stationary polar layer and the air gap. Post-polarization, the change in FENG's internal dipole moment induces a charge distribution on its upper and lower electrodes, where the upper surface and the lower surface exhibit charge density σ . Based on this charge distribution, the total charge of FENG is:⁶²

$$Q(t) = Q_0 \left(\frac{\varepsilon_g d_e}{\varepsilon_r d_{air_0}} + 1 \right) - Q_0 \frac{\varepsilon_g d_e}{\varepsilon_r d_{air_0}} e^{-\frac{1}{RS\varepsilon_0} \left[\frac{d_e}{\varepsilon_r} t + \int_0^t \frac{d_{air}(m)}{\varepsilon_g} dm \right]} - Q_0 \left(\frac{\varepsilon_g d_e}{\varepsilon_r d_{air_0}} + 1 \right) \frac{d_e}{RS\varepsilon_0 \varepsilon_r} e^{-\frac{1}{RS\varepsilon_0} \left[\frac{d_e}{\varepsilon_r} t + \int_0^t \frac{d_{air}(m)}{\varepsilon_g} dm \right]} \times \int_0^t e^{\frac{1}{RS\varepsilon_0} \left[\frac{d_e}{\varepsilon_r} t + \int_0^m \frac{d_{air}(x)}{\varepsilon_g} dx \right]} dm \quad (1)$$

where, ε_0 signifies the permittivity of a vacuum, ε_r and ε_g correspond to the relative permittivity of the dielectric material and the gas, respectively. Moreover, d_e pertains to the overall thickness of the dielectric, whereas d_{air} refers to the comprehensive thickness of the gas layer, and d_{air_0} is specifically used to indicate the thickness of the gas layer in its initial state. Q_0 denotes the quantity of charge following the polarization of the dielectric.

The piezoelectric properties of FENG can be measured by the transverse piezoelectric coefficient d_{33} and the longitudinal piezoelectric coefficient d_{31} . Moreover, $d_{33} = Q/F$, where Q represents the redistributed and accumulated surface charges resulting from mechanical deformation, and F denotes the vertically applied force on the FENG. $d_{31} = Qt/F'l_0$, where t is the thickness of the FENG, l_0 is the length of the electrode, and F' is the force applied to the electrode area of the FENG. The larger the piezoelectric coefficient, the better the performance of the FENG. In quasi-static and dynamic measurements, elements of the piezoelectric tensor d are typically determined. Each component of the piezoelectric tensor represents the piezoelectric response of the material in different directions. The piezoelectric tensor of cell polymers is similar to that of ferroelectric polymers, consisting of shear coefficients d_{15} and d_{24} , longitudinal piezoelectric coefficients d_{31} and d_{32} , and transverse piezoelectric coefficients d_{33} . The piezoelectric tensor d can be expressed as:⁶³

$$d = \frac{\partial D_i}{\partial T_a} = \frac{\partial S_a}{\partial E_i} = \begin{pmatrix} 0 & 0 & 0 & 0 & d_{15} & 0 \\ 0 & 0 & 0 & d_{24} & 0 & 0 \\ d_{31} & d_{32} & d_{33} & 0 & 0 & 0 \end{pmatrix} \quad (2)$$

where, D_i and E_i represent displacement and electric field vectors, respectively, while S_a and T_a denote mechanical strain and stress tensors. In dynamic measurements, the dynamic piezoelectric coefficient can also be expressed as $d_{33} = Q/ma$, where m is the mass of the object acting on the FENG surface, and a is the acceleration of the object moving on the FENG surface. Moreover, the thickness of FENG and the surface charge

also affect the performance of the ferroelectric. Surface charge Q' can be expressed as: $Q' \approx V_s S \varepsilon_{rp} \varepsilon_0 / t$,⁶⁴ where V_s for surface potential, and S for the effective area of the ferroelectric film. ε_{rp} and ε_0 represent the dielectric constant and the vacuum dielectric constant, respectively. Therefore, as shown in Table 1, for FENGs fabricated from materials such as PP or polydimethylsiloxane (PDMS), a thinner FENG results in a larger surface charge, thereby enhancing the piezoelectric performance.

With its distinct bidirectional energy transformation mechanism and inherent properties, FENG showcases immense potential and value in the realm of medical monitoring applications. The creation of FENG marks a notable progression in the sphere of piezoelectric energy conversion, the roots of which can be traced back to an early patent.⁶⁹ Unlike electrodipoles, FENG remains in an entirely non-polar state when unprocessed, only revealing polarity after the internal voids are charged by the dielectric barrier.^{54,70,71} While FENG exhibits behavioral patterns akin to the well-known traditional piezoelectric polymers or piezoelectric composite materials on a macroscopic scale, significant disparities exist in their microscopic structures and operational principles. Compared to conventional piezoelectric materials, ferroelectret materials are unique due to their higher

TABLE 1 Piezoelectric properties of FENG with different thicknesses.

Materials	Thickness (μm)	d_{33}	Year	References
PTFE	200	2.4 pC N ⁻¹	2017	64
PTFE	50	8 pC N ⁻¹	2017	64
PP	100	175 pC N ⁻¹	2014	65
PP	80	400 pC N ⁻¹	2020	66
PDMS	3000	206 pC N ⁻¹	2021	67
PDMS	200	520 pC N ⁻¹	2019	68

Abbreviations: PDMS, polydimethylsiloxane; PP, polypropylene; PTFE, polytetrafluorethylene.

piezoelectric coefficients and lower Young's modulus. The Young's modulus of FENG can be obtained by:⁷²

$$Y = \frac{F/A}{l'/L} = \frac{FL}{l'tw} \quad (3)$$

where A is the cross-sectional area F is the force applied in the length direction of the strip, and l' is the dynamic deflection, L and w are the length and thickness of the FENG layer, respectively. The smaller the Young's modulus, the softer the FENG, making it easier to fit human body and better used for body function testing.

These traits are derived from their charged cellular structures, rendering ferroelectret materials more responsive to stress variations and capable of effectively storing charge. Owing to this unique physical mechanism, FENG amalgamates the high piezoelectric performance demonstrated by inorganic piezoelectric compounds (such as lead zirconate titanate piezoelectric ceramic (PZT)) with the flexible film structure inherent to organic piezoelectric polymers (e.g., polyvinylidene fluoride (PVDF)). The FENG, when affixed to the human skin or implanted within the body, can efficaciously monitor human movement via the wireless sensing mechanism as depicted in Figure 2. Specifically, the FENG, regardless of whether it is implanted internally or externally adhered to the skin, can detect pressure fluctuations originating from both within and outside the body with high sensitivity. These pressure variations result in the deformation of the FENG, subsequently generating electrical current or voltage. Conversely, alterations in the potential on the FENG surface can also influence human physiological movements. Therefore, the bidirectional conversion of mechanical and electrical

energy in the human body is facilitated through the application of FENG's direct and reverse piezoelectric effects.

2.2 | Preparation method

The fabrication of FENG involves a variety of methodologies, with the primary ones being the sol-gel method, vapor deposition, and electrochemical deposition. These generators typically exhibit a sandwich-like metal-insulator-metal (MIM) structure, a feature that enables large-scale manufacturing of FENG. Table 2 details the materials frequently used in FENG fabrication, including PP,^{51,77,78} irradiation cross-linked polypropylene (IXPP),⁷⁹ polypropylene ferroelectret (PPEE), Teflon AF, Ecoflex, polyethylene terephthalate (PET), ethylene vinyl acetate copolymer (EVA), polytetrafluorethylene (PTFE) and expanded polytetrafluorethylene (ePTFE),^{80–83} among others. Additionally, FENGs can be designed in various forms; for example, an array made of ferroelectret sensor unit, or a sandwich structure constructed from 3D printed TPU meshes and fluorinated ethylene propylene (FEP) layers, as depicted in Figure 3A,B, respectively. These materials, given their remarkable malleability and superior processing capability, can be shaped into thin, robust films. As shown in Figure 3C, laser scribing can be employed to produce laser-induced graphene (LIG) patterns, and a multilayer porous structure of FENG can be fabricated by dropping a vigorously mixed solution of Ecoflex and anhydrous ethanol, yielding a FENG with exceptional flexibility, stretchability, and both longitudinal and transverse piezoelectric effects.

Figure 3D illustrates a common method of fabricating FENG using high-pressure gas injection technology

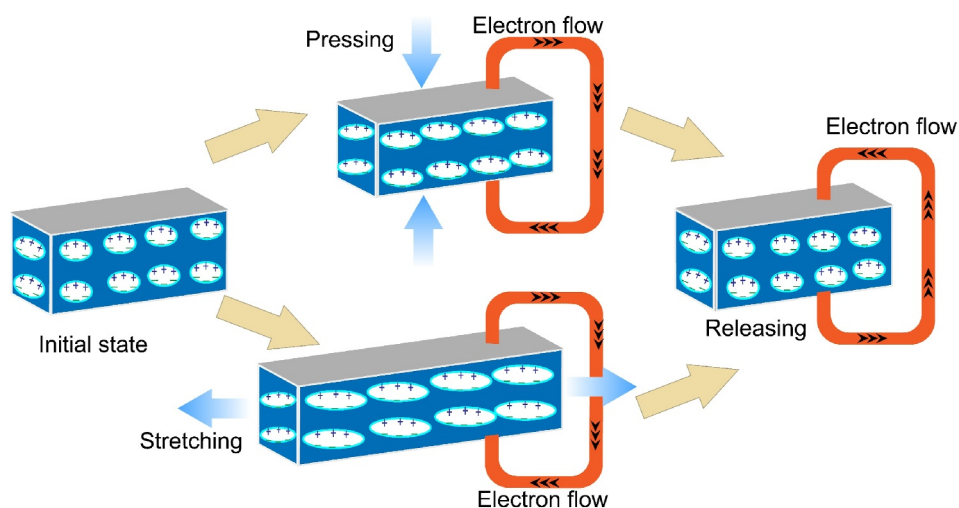


FIGURE 2 Principles of ferroelectret nanogenerators.

TABLE 2 Application, manufacturing methods, and charging methods of various FENGs.

Materials	Manufacturing methods	Charging methods	d_{33}	Application	Year	References
PPEE	Heat curing	Plasma charging	400 pC N ⁻¹	Energy harvesting	2016	51
LIG/Ecoflex/ethanol	Laser ablation and heat curing	Corona charging	120 pC N ⁻¹	Sports monitoring	2024	73
Au/parylene/ Teflon AF	Chemical vapor deposition	Corona charging	500 pC N ⁻¹	Respiration monitoring	2021	74
PP	Gas diffuses	Corona charging	309 pC N ⁻¹	Sports monitoring	2018	53
PP	Thermodilatometry	Corona charging	510 pC N ⁻¹	Pulse monitoring	2019	75
FEP/Ecoflex	Solvent casting	Corona charging	–	Sports monitoring	2019	47
PET/EVA	Hot-pressing	Corona charging	6300 pC N ⁻¹	Energy harvesting	2017	76

Abbreviations: EVA, ethylene vinyl acetate copolymer; FEP, fluorinated ethylene propylene; LIG, laser-induced graphene; PET, polyethylene terephthalate; PP, polypropylene; PPEE, polypropylene ferroelectret.

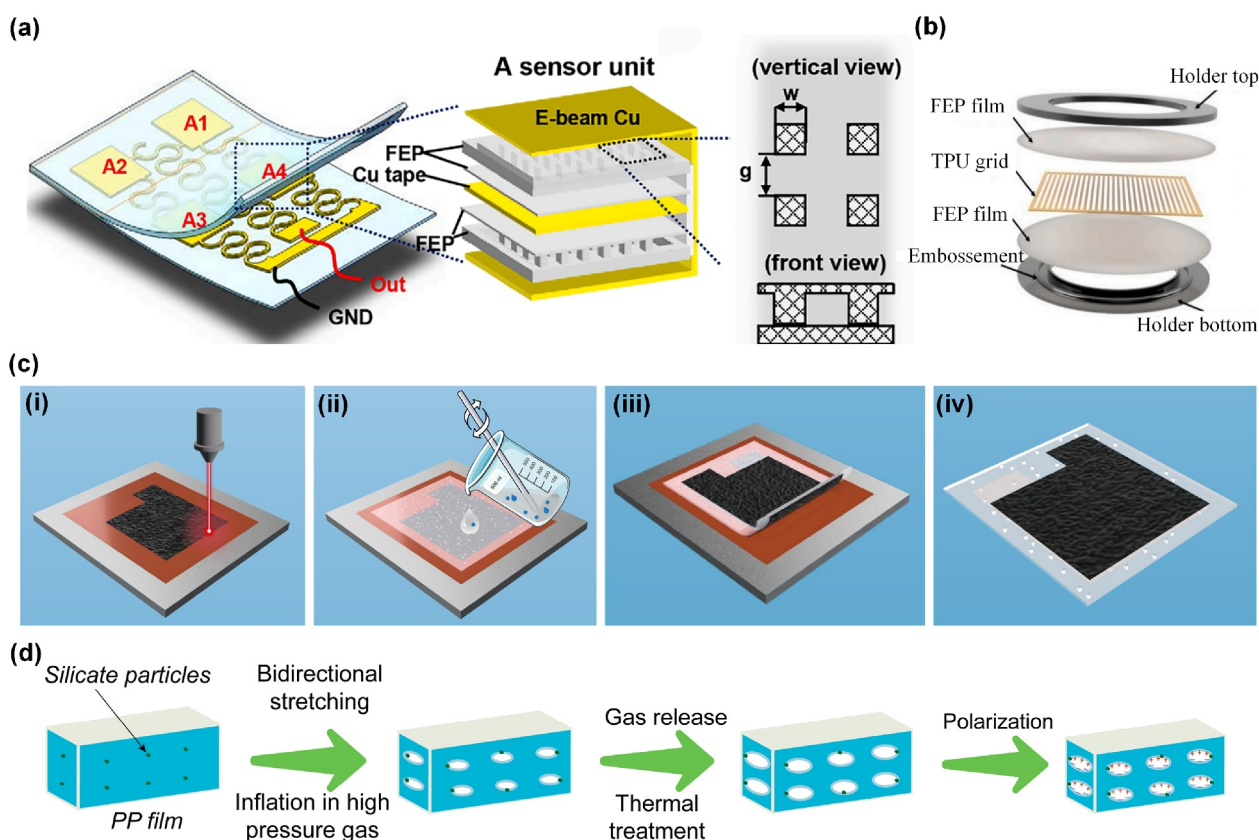


FIGURE 3 Preparation of FENG. (A) Schematic diagram of the proposed device containing four sensor units (A1~A4). Reproduced with permission.⁸⁴ Copyright 2024, Elsevier. (B) Sandwich structure consisting of a 3D-printed TPU grid and fluorinated ethylene propylene layers. Reproduced under terms of the CC-BY license.⁸⁵ Copyright 2021, published by MDPI. (C) Schematic diagram of the HS-FENG fabrication process, including: (i) the creation of laser-induced graphene patterns via CO₂ laser scribing of polyimide film, (ii) drop casting of a vigorously mixed solution of Ecoflex and anhydrous ethanol, (iii) the hot curing of the entire HS-FENG followed by its removal from the substrate, and (iv) a depiction of the HS-FENG prior to polarization, showcasing its multi-layer structure containing voids. Reproduced with permission.⁷³ Copyright 2024, Elsevier. (D) A typical method of preparing FENG using high-pressure gas injection combined with small inorganic particles. Reproduced under terms of the CC-BY license.⁸⁶ Copyright 2023, published by MDPI. FENG, ferroelectret nanogenerators.

coupled with inorganic micro-particles, utilizing a PP film embedded with minuscule silicate particles.^{63,87} When the film is stretched in two orthogonal directions, these inorganic particles serve as stress concentration points or micro-crack initiators, enhancing the stress distribution in the chosen materials (especially PP films). In this process, high-pressure nitrogen gas (e.g., 5 MPa) is pumped into the PP film, equalizing the pressure within the film's voids and the external pressure. The external gas pressure is then abruptly relieved, leading to the expansion of the voids within the PP film. A heat treatment (e.g., 4100°C) is then applied to solidify these expanded voids. Finally, a layer of liquid metal is deposited onto the surface using sputtering technology.

To utilize FENG in the medical field, it occasionally necessitates its implantation within human body for regular functioning. This demands encapsulation material to be non-toxic, harmless, and exhibit superior biocompatibility. Studies have corroborated that the application of a biocompatible polyimide (PI) film to encapsulate flexible devices can effectively prevent device's functional components from body fluids, thereby markedly mitigating the risk of failure or eliciting immune responses in bio-implant applications.⁸⁸ Moreover, in vitro examinations on human epidermal keratinocytes have demonstrated the non-toxicity of PI film layers.⁸⁹ Given its high flexibility and tensile strength, PI film emerges as a promising encapsulation material for FENG.

3 | APPLICATION OF FENG IN MEDICAL FIELD

FENG is a proven, efficient, lightweight, reliable, and highly compatible solution, offering economical advantages across various sectors, including mechanical energy collection and self-powered sensing.^{47,90–94} As a novel technology in the realm of electromechanical energy conversion, FENG has been effectively harnessed to generate power from a diverse range of biomechanical movements, including but not limited to arm movements, respiration, sound vibrations, walking, heartbeat, pulse wave, blood flow, vascular pressure, and gastric peristalsis.^{14,95} Beyond its role as an energy converter, FENG also functions as a bio-information medium, hence bridging communication between humans and devices, systems, or machines. This has enabled the transmission of bio-information in devices such as keyboards, microphones, speakers, and fitness trackers.^{96,97}

In the context of the Internet of Things (IoT), FENG's potential in biomedical engineering is set to be fully exploited. The proliferation of implantable and wearable

electronic products, along with biosensors, is crucial for personalized healthcare and Human-Machine Interaction (HMI). The absence of a battery (i.e., self-powered) and the flexibility of FENG significantly enhance the usability of various devices in biomedical engineering applications. Consequently, FENG's energy conversion capability provides an efficient method for effectively harnessing biomechanical energy from the human body or the environment. Apart from generating biomechanical energy and serving as a biosensor, FENG's inherent properties also allow for bidirectional energy conversion between electrical and mechanical power sources, thereby expanding its application range in bioelectronics. FENG has shown immense potential in biomedical devices and has been successfully integrated into actual clinical treatments for various patients.^{98–100}

In conclusion, FENG, a breakthrough in nanogenerators technology, amalgamates self-power supply, actuator, and sensor capabilities. It showcases exceptional ease of fabrication, unparalleled stretchability, precise manufacturing abilities, and a high efficiency in energy conversion between electricity and mechanics. Its widespread use in the medical sector, notably in areas like blood circulation monitoring, respiratory system evaluation, muscle movement analysis, and neurological reflex testing, underscores its vast potential and the forward direction it sets for bioelectronic technology.

3.1 | Applications in the blood circulation

The heart, a critical organ responsible for blood circulation and monitoring, represents a significant area of study within circulatory research. Piezoelectric ultrasonic energy harvesting (PUEH) technology, with its potential in supplying wireless power to bio-implantable devices, is of particular interest. To enhance cardiac activity monitoring, researchers have innovatively developed a flexible PUEH array based on thin film,¹² as depicted in Figure 4A. This array, utilizing ultrasonic waves, can wirelessly charge bio-implantable devices and operate efficiently in diverse curved environments. The device's core components include a set of lightweight, high-performance piezoelectric composites, known as PZT/epoxy resins, which form the fundamental unit for energy harvesting. The design also incorporates a double-layer wavy copper wire for flexible electrical interconnection, and an elastomer membrane for protective encapsulation to ensure the structure's overall flexibility and durability. This ingenious design not only guarantees effective polarization of the piezoelectric components, but also enhances the device's flexibility via the wavy copper wire.

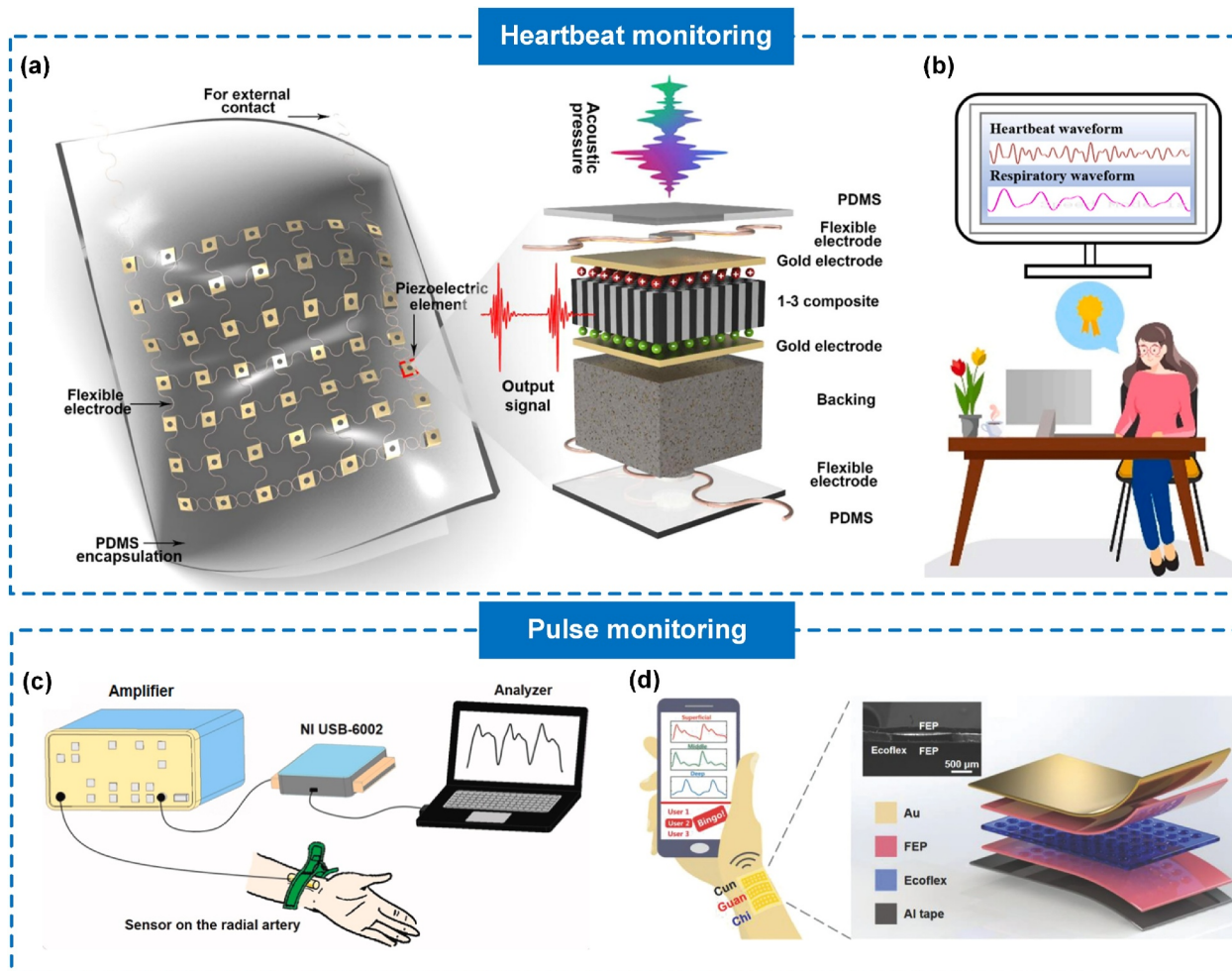


FIGURE 4 Applications of ferroelectret nanogenerators in the blood circulation. (A) Schematic illustration of the Piezoelectric ultrasonic energy harvesting structure. An exploded view further elucidates each component within a piezoelectric element. Reproduced with permission.¹² Copyright 2019, Elsevier. (B) Schematic illustration of the health management platform for the office sedentary crowd. Reproduced with permission.¹⁰¹ Copyright 2023, Elsevier. (C) A schematic diagram delineates the experimental setup. Reproduced with permission.⁷⁵ Copyright 2019, Elsevier. (D) The concept of the m-Health system, based on the pulse-sensing system, and the inset (upper left corner) reveal a cross-sectional SEM image of the sandwich-structured piezoelectret film. Reproduced with permission.⁴³ Copyright 2018, Wiley-VCH Verlag.

The PUEH device can consistently deliver a stable power output, fueled by ultrasonic waves. To evaluate the implant's performance at varying tissue depths, the experiment simulated conditions with pork tissue of different thicknesses (serving as a biological tissue substitute) placed between the transmitter and the PUEH device. The findings revealed no significant output signal attenuation when the pork tissue thickness increased from 0 to 14 mm. Specifically, at a tissue thickness of 14 mm, the generated voltage was 0.91 Vpp, equivalent to 85% of the voltage (1.07 Vpp) without a tissue barrier. Concurrently, the corresponding current signals were 3 and 2.5 A, respectively, indicating that the PUEH device can maintain high energy conversion efficiency at different tissue depths.

By fine-tuning the frequency of the ultrasound, the sound reflection and absorption instigated by the impedance mismatch between the tissue and the heart can be significantly minimized. This optimization enhances the transmission efficiency of the ultrasound energy.¹⁰² The proposed flexible PUEH device, therefore, not only possesses the capacity to harness energy within biological tissues but can also accomplish embedded energy extraction. This innovation paves the way for wireless power supply in various bio-implant systems such as high-density neural stimulators,¹⁰³ cardiac defibrillators,¹⁰⁴ and pacemakers.¹⁰⁵ In related work, a different research group¹⁰⁶ has engineered a micro-helical piezoelectric energy harvester, encapsulated using microfabrication technology. This apparatus has the

ability to directly harness energy from alterations in cardiac blood pressure, thereby making it an enduring micro-energy source ideal for implantable medical devices such as lead-free pacemakers. The researchers conceived an ultra-thin and ultra-flexible electro-deposited microrib tube package for this device, allowing it to seamlessly adapt to the heart's dynamic changes and demonstrating robust adaptability to fluctuations in heartbeat frequency. Experimental results indicate that this device can generate an energy density of $3 \mu\text{J cm}^{-3}$ with each heartbeat, further substantiating its potential application in the realm of implantable medicine.

In the realm of blood circulation monitoring, a pivotal study by FENG emphasizes pulse monitoring. This technique is a fundamental method deployed by trained physicians in Traditional Chinese Medicine (TCM) to pinpoint potential health problems. The health condition of organs and diverse body parts is mirrored in the pulse, which is influenced by blood circulation throughout the body. This is substantiated by the contraction and relaxation activities of the heart, coupled with the dynamic pressure alterations when blood courses through the arteries.^{107–109} Nonetheless, the ECG, widely employed in current clinical practice, is somewhat restricted for daily monitoring due to its substantial size and portability issues. To facilitate long-term, continuous pulse monitoring, and to ensure the accuracy and dependability of the monitored data, numerous research teams are committed to the development of highly sensitive, portable, self-driven pulse sensors. These sensors are paramount for the advancement of mobile medical and telemedicine technologies.

In recent years, an array of pulse wave measurement sensors have been introduced, including acoustic sensors, piezoelectric sensors, image sensors, capacitive sensors, and piezoelectric poling sensors. Notably, the piezoelectric poling sensor, with its high equivalent piezoelectric coefficient d_{33} , exhibits exceptional sensitivity. The piezoelectric attributes of this sensor stem from the charge stored in the material voids, enabling the sensor to operate independently of external power sources, as demonstrated in energy collector applications. Moreover, its low elastic modulus characteristic renders it flexible in configuration, making it suitable for a variety of potential applications such as tactile perception and wearable sensing, as outlined in Table 3.

Porous polypropylene, as a classic ferroelectret material, demonstrates enormous application potential in the field of self-powered wearable sensing due to its high equivalent piezoelectric coefficient, excellent stability, lightness, and ease of manufacture. This is especially the case when it comes to detecting weak physiological pressure signals such as pulse waves and blood pressure.

Researchers have recently reported a highly sensitive, flexible thin film pressure sensor with a folded structure, assembled from IXPP piezoelectric electret films. This sensor displays piezoelectric-like dynamic electrical output characteristics. The simple folding structure is integrally designed into the device, providing piezoelectric thin film sensors with advantages like increased sensitivity, high signal-to-noise ratio, and excellent anti-interference capability.¹⁰¹ The sensor can achieve a linear pressure sensitivity of 92 pC/kPa within a broad pressure range of 0.6–62.4 kPa. It has a pressure resolution of 0.98% and a fast response/recovery time of 40 ms. A health management platform has been developed for sedentary individuals. This platform includes an intelligent seat cushion with piezoelectric electret sensors inserted, a data acquisition/front-end processing module, and a signal analysis display. The sensor is embedded into the sponge pad, which acts as the medium between the human body and sensors. This allows individuals to monitor their HR and respiratory rate in real-time while seated, without the need to wear any additional devices. This design, as shown in Figure 4B, further showcases its extensive potential applications in wearable medical monitoring and clinical diagnosis.

To enhance the monitoring of pulse conditions effectively, an innovative flexible porous polypropylene piezoelectric film has been utilized as a sensing medium by researchers.⁷⁵ This film employs a wooden cylinder as a substrate, with the intention to emulate the traditional pulse-taking methodology of TCM, and it captures pulse characteristics as depicted in Figure 4C. The results indicate that the approximate entropy (ApEn) value can serve as a powerful tool for assessing an individual's health status. Specifically, a decrease in physiological adaptability was observed in correspondence with the deterioration of the health status of volunteers, which was concurrently associated with a decrease in the ApEn value. Consequently, this piezoelectric sensing system not only streamlines the operation process, facilitating the attainment of low-distortion pulse data for comprehensive analysis, but also validates its practicality in pulse condition monitoring. Moreover, researchers have also employed a piezoelectric pulse wave sensing system based on the FEP/Ecoflex/FEP sandwich structure for piezoelectric detection (refer to Figure 4D). This system is capable of accurately identifying and tracking the subtle vibration patterns of the human radial artery, and its exceptional precision and stability have been substantiated.⁴³ This not only expands the technical methods of pulse condition monitoring, but also further augments the application potential of piezoelectric sensing technology in the modernization process of TCM diagnosis.

TABLE 3 Applications of FENG in the field of medical engineering.

Materials	Structures	Sites	Sensitivity	Monitoring objects	Year	References
PP	Cellular form	Hand and wrist	–	Muscle contraction	2018	53
PP	Cellular form	Wrist	–	Blood pressure and heartbeat	2019	75
PP	Cellular form	Wrist	0.21 V kPa ⁻¹	Bending and relaxation movements	2022	110
PP	Cellular form	Brain	–	Brain stress	2023	111
PP	Cellular form	Wrist, arm and neck	–	Heartbeat	2015	112
FEP	Parallel-tunnel	Throat and elbow	1600 pC mm ⁻¹	Muscle contraction	2022	96
FEP/EVA	Sheath-core configuration	Body	18.98 V kPa ⁻¹	Sleep	2018	113
FEP/Ecoflex	Sandwich composite	Wrist and arm	1.45 V kPa ⁻¹	Blood pressure	2019	4
FEP/Ecoflex	Sandwich composite	Wrist	32.6 nA kPa ⁻¹	Blood pressure and heartbeat	2018	43
IXPP	Fold	Sitting motion	92 pC kPa ⁻¹	Respiration and heartbeat	2023	101
PZT	Fold	Heart	–	Heartbeat	2019	12
LIG/Ecoflex/ethanol	Sandwich composite	Muscles and joints	–	Sport	2024	73
Teflon AF	Sandwich composite	Nose	–	Respiration	2022	74
–	Cellular form	Body	–	Sleep	2020	99
Fluorocarbon	Sandwich composite	Waist	31.4 μ W cm ⁻²	Respiration	2016	114
PET	Fold	Brain	–	Brain stimulation	2015	115
PVB/PDMS	Cylindrical shape	Nerve	–	Nerve stimulation	2022	116
PDMS	Cellular form	Wrist	10 mV Pa ⁻¹	Heartbeat and blood pressure	2014	117

Abbreviations: EVA, ethylene vinyl acetate copolymer; FEP, fluorinated ethylene propylene; IXPP, irradiation cross-linked polypropylene; LIG, laser-induced graphene; PDMS, polydimethylsiloxane. PET, polyethylene terephthalate; PP, polypropylene; PVB, polyvinyl butyral; PZT, zirconate titanate piezoelectric ceramic.

FENG's research has highlighted a crucial application in the vascular domain, specifically, the use of nano-robots developed by FENG to harness energy from blood for the facilitation of vascular dredging. Thrombotic diseases are infamous for their swift onset,¹¹⁸ particularly cerebral thrombosis, where an estimated 1.9 million neurons perish per minute in patients, resulting in a rapid deterioration of neurological function. The absence of timely thrombolytic treatment could potentially lead to paralysis or life-threatening conditions.¹¹⁹ Currently, the primary treatment methods for thrombotic diseases are surgical thrombectomy¹²⁰ and drug thrombolysis.¹²¹ However, surgical thrombectomy not only poses high risks, but also inflicts substantial physical discomfort on patients. Large doses of thrombolytic drugs, on the other hand, may result in side effects like bleeding, especially pronounced in the elderly population with a high prevalence of cardiovascular diseases.

3.2 | Applications in the respiratory system

Respiratory-related signals serve as critical indicators for assessing health status and have been extensively explored through various methods. These methods predominantly involve the use of Doppler radar-based systems¹²² and thermopile/calorimetric respiratory sensing systems.¹²³ However, these methods' practicality is hindered by their substantial size, reliance on external power sources, and potential discomfort for users, thus limiting their widespread application in the HMI sector.

Addressing these challenges, FENG has been employed to harvest respiratory flow mechanical energy from various body parts and extract essential data, such as respiratory rate, depth, and pattern (including swallowing, deep breathing, slow breathing, and rapid breathing). Chen et al. presented a novel, non-contact

sleep monitoring system.¹¹³ The core component of this system is a self-powered sensing strip, which is anchored on a flexible, hollow microstructure (HM) based FENG. As depicted in Figure 5A, the external sheath of this HM is composed of FEP/silver flakes, while the inner core is made up of EVA/silver flakes. Ingeniously, this system is designed to be situated on the bed rather than worn on the body, as shown in Figure 5B. As illustrated in Figure 5C, this sensor strip can generate an open-circuit voltage of up to 110 V under a pressure of 5 N and a frequency of 5 Hz, demonstrating exceptional stability even after 5000 test cycles. Further investigation revealed the sensor strip's sensitivity in low pressure (< 1 kPa), medium pressure (1.6–10 kPa), and high pressure (10–40 kPa) regions to be 18.98 V/kPa, 0.77 V/kPa, and 0.25 V/kPa respectively, showcasing superior performance across diverse pressure ranges.

In the field of ballistocardiography (BCG) measurements, it is possible to assess stress levels and recovery status through HR and HR variability (HRV) measurements. In 2020, a study was conducted where researchers used ECG equipment as a reference to validate the accuracy of quantified sleep (QS) in measuring HR and HRV during sleep.⁹⁹ In this study, a QS device (refer to Figure 5D), measuring $6 \times 55 \text{ cm}^2$ and placed under the mattress, was used to record nighttime HR, HRV, respiration, and other body movements, as depicted in Figure 5E. Once the participants fell asleep, the QS device would automatically begin measurements and stop when they awoke.

The QS device not only finds extensive application in the health monitoring of athletes and sedentary individuals, but also innovatively converts physiological data into actionable insights. The QS device works in conjunction with a sophisticated cloud-based analysis platform, allowing for precise measurements of BCG, respiration, body movement, HR, and respiration rate (RR).⁹⁸ The system also performs a comprehensive analysis of HR variability and provides a summary of the previous night's activities, including estimates of sleep stages and indicators of sleep quality. A clinical sleep study was conducted in the Pirkanmaa Hospital District in Tampere, Finland, which utilized the QS device. The data from this study revealed that the HR and RR values derived from the QS device closely matched the values reported in the existing literature. Consequently, this high-precision, non-contact FENG device demonstrates significant value as it allows for the convenient monitoring of vital signs without causing any discomfort to the subjects.

Researchers have recently investigated an innovative sandwich composite fluorocarbon (SCF) thin film system¹¹⁴ that exhibits the ability to detect human

respiratory signals. The SCF thin film system, when compared with traditional dielectric materials such as dense vinylidene fluoride or fibrous polytetrafluoroethylene thin films of identical thickness, can retain more residual charges, thereby showcasing superior performance. A dielectric generator, based on the SCF thin film and under a load resistance of 40 M Ω , can achieve a maximum peak power density of 31.4 $\mu\text{W cm}^{-2}$. Additionally, the implementation of a self-powered respiratory monitoring sensor based on the dielectric generator has been demonstrated by researchers. This sensor can be integrated with a waist belt to detect muscle movements instigated by human respiration. Notably, the generated positive and negative current peaks are a result of respective inhalation and exhalation movements, and not background noise. This observation further attests to the sensor's accuracy and reliability in monitoring respiration. The sensor also possesses the capability to measure varying respiratory states, thereby offering new technological means for monitoring respiratory health.

In parallel, researchers have also created an intelligent mask embedded with an ultra-thin, self-powered pressure sensor,⁷⁴ as depicted in Figure 5F. Constructed from two layers of Au/PTFE/Teflon AF film bonded by Au-Au, the sensor's structure is illustrated in Figure 5G. A high-density super-expanded dipole within the device endows the sensor with exceptional quasi-piezoelectric performance, enabling the wireless monitoring of respiratory information. This self-powered pressure sensor is not only the thinnest and lightest electrostatic pressure sensor, with a total compression thickness of just 5.5 μm and a weight of merely 4.5 mg, but also capable of outputting a peak voltage of up to 10 V under airflow stimulation. This unique feature aids in the miniaturization of circuits and reduction of power consumption. To support wireless operation and user monitoring, researchers have incorporated a compact readout circuit into the sensor and designed a mobile application for interaction with the intelligent mask. As Figure 5H demonstrates, the intelligent mask exhibits excellent output stability, continuously monitoring and analyzing respiratory conditions, including normal breathing, rapid breathing, coughing, and breath-holding. It thus provides a new approach for respiratory health monitoring.

3.3 | Applications in the muscle movement

The increasing demand for innovative intelligent sports technology necessitates sensors capable of real-time monitoring of individual physiological movement information, including physiological responses and muscle

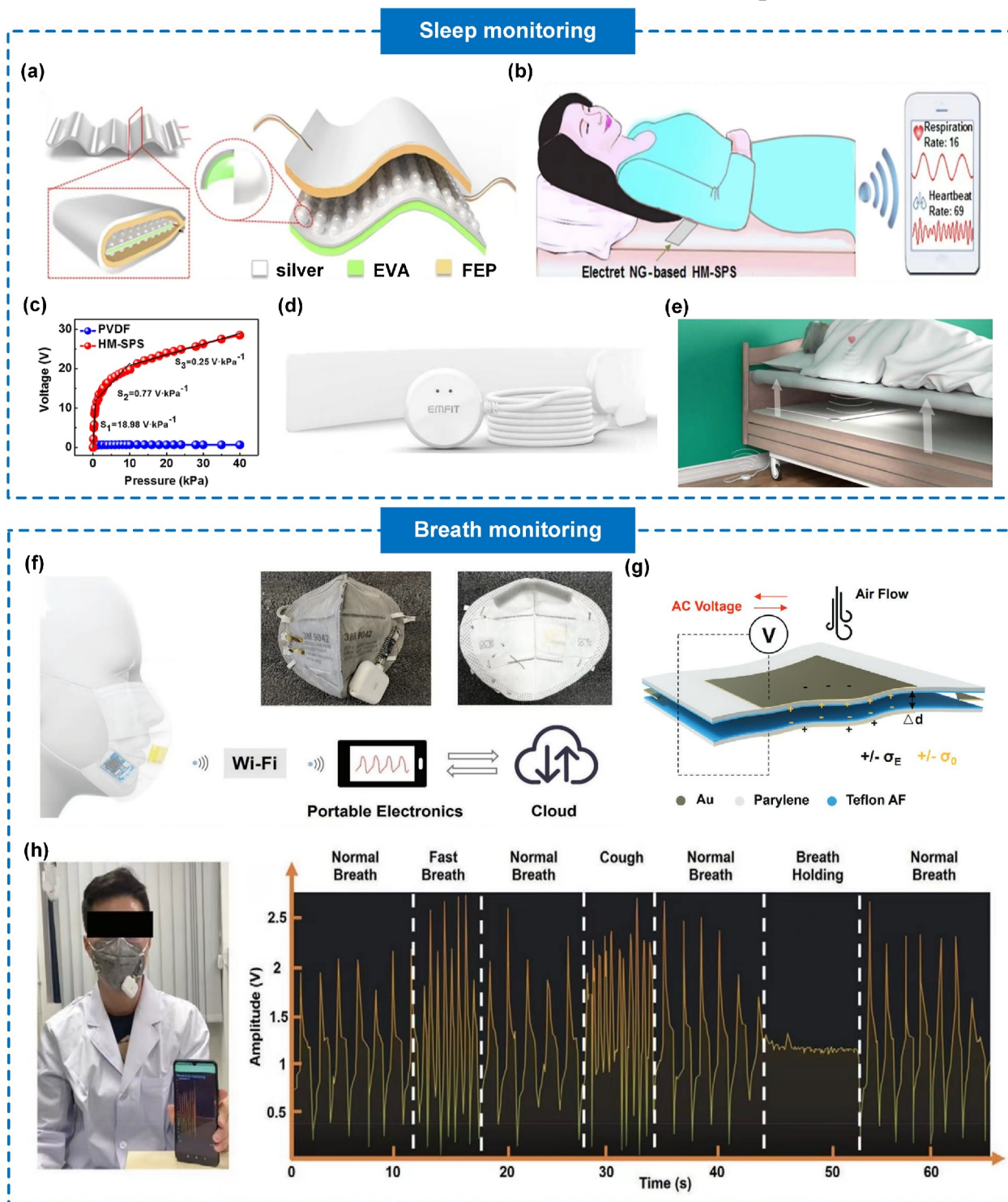


FIGURE 5 Applications of FENG in the respiratory system. (A) The schematic structure of the HM sensor, including an overall view, an oblique section view, and an expanded view, (B) the schematic representation of the non-contact sleep monitoring system and (C) comparisons of the pressure response curve between the HM sensor and a commercial PVDF pressure sensor. Reproduced with permission.¹¹³ Copyright 2018, ACS. (D) The depiction of the placement for the Emfit quantified sleep sensor, and (E) the schematic illustration of the FENG sensor with a processing unit streaming data to the cloud via the Internet. Reproduced under terms of the CC-BY license.⁹⁹ Copyright 2020, published by JMIR Publications. (F) The smart mask equipped with an ultra-thin, self-powered pressure sensor, (G) constructed from a two-layer Au/poly-p-xylylene/Teflon AF film secured by an Au-Au bond, and (H) demonstrating exceptional output stability. Reproduced with permission.⁷⁴ Copyright 2022, Wiley-Blackwell. FENG, ferroelectret nanogenerators; HM, hollow microstructure.

activity status. Researchers have ingeniously combined Ecoflex, anhydrous ethanol, and LIG to design an HS-FENG,⁷³ as illustrated in Figure 6A. This design not only bestows HS-FENG with significant longitudinal and transverse piezoelectric effects, but also establishes a novel pathway for the development of skin-mounted intelligent movement sensors and robotic electronic skin. Notably, HS-FENG can be utilized to develop intelligent kinesiology tape capable of accurately capturing real-time movement signals of human muscles and joints, offering valuable feedback for athletes and fitness enthusiasts. This technology, in addition to aiding posture correction and training effect assessment, presents numerous advantages such as self-powering, high stretchability, flexibility, biodegradability, and cost-effectiveness.

Muscle contraction generates radial force distribution on the muscle surface due to volume changes, thereby creating pressure in the radial direction. As a result, body movements can be recorded using force myography (FMG) technology. To augment the application potential of FMG technology, researchers have refined the preparation process of the PP porous membrane, developing an FMG recording system based on a ferroelectric sensor,⁵³ as depicted in Figure 6B. Employing this system, they successfully detected varying movement patterns of a healthy individual's hand and wrist, as shown in Figure 6C. By analyzing and identifying FMG patterns using linear discriminant analysis and artificial neural network algorithms, they achieved an average movement classification accuracy of up to 96.1% and 94.8%, respectively. This accomplishment indicates extensive application potential in the realms of biomedical engineering and structural health monitoring.

Ferroelectret is a kind of charged polymer with a pore structure, it can produce a significant dipole moment in the direction of material thickness under stress. Capitalizing on this feature, researchers have developed a wearable textile substrate tactile sensor based on PP.¹¹⁰ This sensor, cleverly adhered to a glove's surface, exhibits a high sensitivity of 0.21 V/kPa within a pressure range of 0.20 kPa. As depicted in Figure 6D, it can accurately detect the palm's bending and relaxation movements, the flexing and extending movements of each finger, and identify the little finger's posture at an output voltage of approximately 400 mV.

In a similar vein, Junwen Zhong and his team designed a flexible sandwich-structured piezoelectric relaxor device that functions both as a sensor and an actuator.⁴⁷ This device comprises two layers of FEP relaxor thin film, an Ecoflex spacer middle layer, and Au and aluminum (Al) electrodes on the top and bottom surfaces. When an alternating current voltage is applied,

the device operates as an actuator driven by electrostatic force, inducing vibrational stimulation and providing tactile feedback to human skin, as shown in Figure 6E. Simultaneously, under mechanical action (such as deformation during human movement), the device can function as a sensor, sensing electric output without an external power source. By integrating the actuator and sensor into a wearable device, it can provide real-time sensing and actuation feedback, and find potential applications in remote tactile communication.

In a bid to further achieve multifunctional observation of human movements, researchers developed a sensor based on parallel tunnel FEP laminated ferroelectret (PTFLF) for multifunctional sensing in wearable devices.⁹⁶ Leveraging the superior charge storage stability of the FEP relaxor thin film and the mechanical compression and tensile properties of the specially designed parallel tunnel void structure, this sensor displays a significant longitudinal piezoelectric effect (with a piezoelectric coefficient of up to 6200 pC/N) and overcomes the lateral piezoelectric effect's shortcomings in most other ferroelectrics. The sensor exhibits a strain sensitivity of 1600 pC/mm and a bending sensitivity of 390 pC/mm. Moreover, the PTFLF sensor can conformally contact different parts of the human body, such as the throat, elbow, wrist, fingers, and knee, as depicted in Figure 6F, thus demonstrating its application potential in wearable devices. This innovative design approach can be applied in fields such as home healthcare and smart mobile medical electronics.

3.4 | Applications in the neural reflex

Throughout the history of neural electrical stimulation research, scholars have consistently exhibited an unwavering spirit of exploration. This domain has seen the advent of numerous innovative electrical stimulation methodologies and devices, all aimed at enhancing the diagnosis and treatment of neurological disorders. However, the frequent occurrence of electrical stimulation inertia due to the application of exogenous electron injection technology not only undermines the therapeutic outcomes, but also incites a range of complications. This significantly impedes the extensive application of electrical stimulation technology within the neural engineering arena.

To mitigate this technical impediment, Kim and his team successfully fabricated a flexible single-crystal Pb(In_{1/2}Nb_{1/2})O₃-Pb(Mg_{1/3}Nb_{2/3})O₃-PbTiO₃ (also known as PIN-PMN-PT or PIMNT) thin film on a PET substrate, as depicted in Figure 7A. They further investigated its potential applicability in deep brain stimulation.¹¹⁵ The

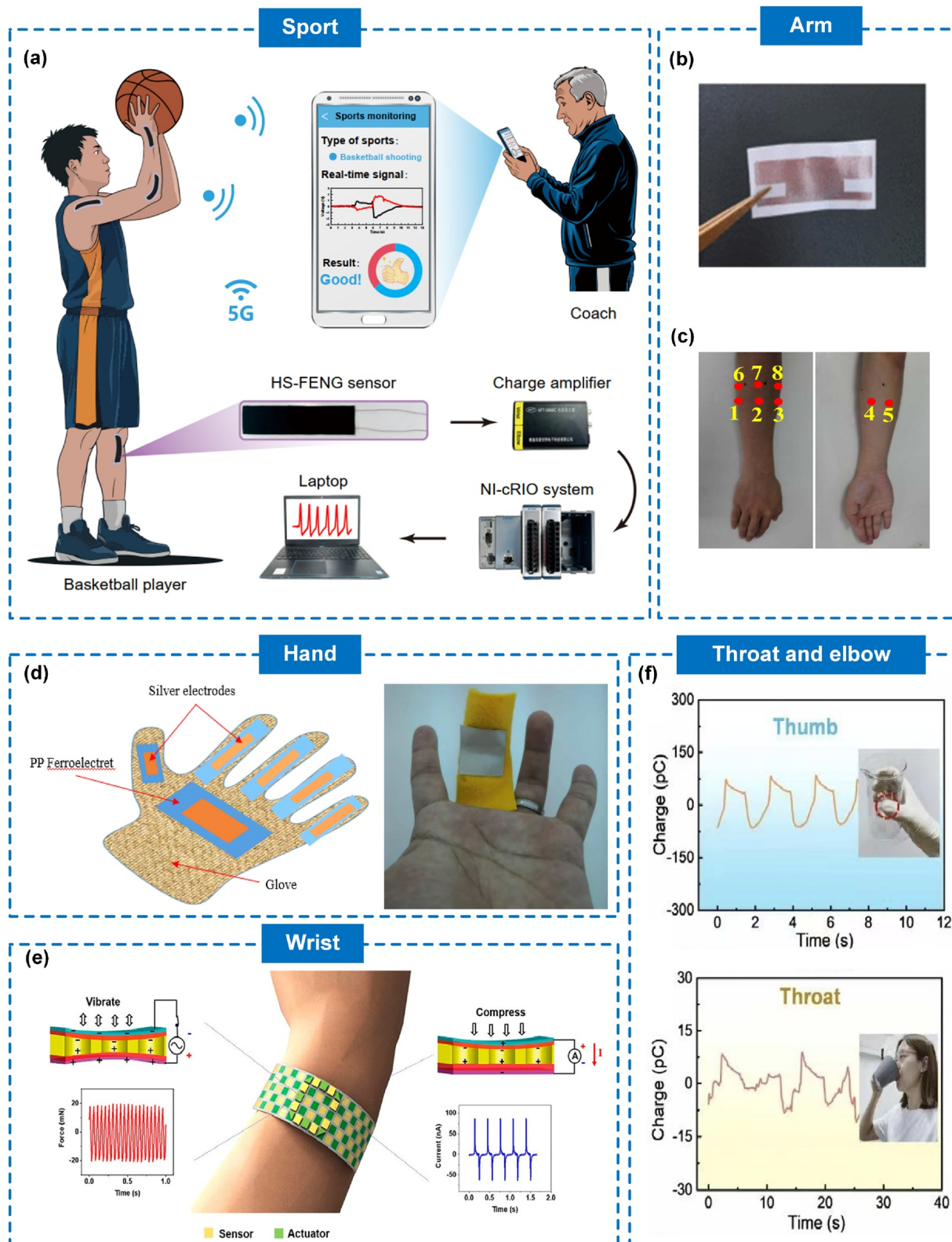


FIGURE 6 Legend on next page.

PIMNT film boasts a mere thickness of 10 μm , while the PET substrate measures 125 μm , together constituting an ultra-thin device. The corresponding simulation model is presented in Figure 7B. Upon exposure to mechanical bending force, this device demonstrates superior electrical performance, achieving a maximum open-circuit voltage of 11 V, a short-circuit current of 283 μA , and an output power reaching 3.11 mW. These parameters are sufficient to power capacitor charging and illuminate up to 120 LED lights. Additionally, when flexed by a finger, the PIMNT nanogenerator can produce a maximum current of 0.57 mA and a power of 0.7 mW. This PIMNT brain stimulator was used to activate the primary motor cortex (M1 area) of a mouse brain, with the electrode implantation sites depicted in Figure 7C,D. Experimental evidence indicates that each flex of the PIMNT nanogenerator can effectively prompt forelimb muscle contraction, resulting in a forward movement of the mouse's right paw by 1.5–2.3 mm. This signifies a significant advancement in achieving direct deep brain stimulation via body movement.

Similarly, Wan et al. developed a multilayer friction FENG by incorporating an array of parallel-aligned pores into the dielectric layer situated between the traditional polar bodies.¹¹⁶ The aim was to enhance the piezoelectric performance of the high-efficiency ultrasonic energy collector (H-UEH), as depicted in Figure 7E. When exposed to an ultrasonic probe with 25 mW/cm², the device demonstrated a peak output power of approximately 13.13 mW and a short-circuit current of roughly 2.2 mA. These readings were taken when the implantation depth was set between 5 and 10 mm, and both exceeded the power threshold necessary for the standard operation of bioelectronic devices and the current threshold needed for nerve stimulation. The viability of an implantable nerve stimulator, predicated on H-UEH, has been substantiated through peripheral nerve stimulation experiments on a rat model. These experiments incorporated stimulation of both the vagus and sciatic nerves, as depicted in Figure 7F. Moreover, the

biocompatibility of the FENG ultrasonic energy harvester was corroborated through subsequent cell culture experiments. The device was also subcutaneously implanted in the rat's back for extended monitoring of its stability and biocompatibility. Post 56 days of implantation, the wound on the rat's back had fully healed, leaving behind only a minor scar. This series of investigations not only underscores the merits of FENG-based ultrasonic energy harvesting technology in terms of biocompatibility and stability, but it also illuminates its wide-ranging application potential in the realm of implantable bioelectronics, such as wireless power supply, drug delivery, or nerve repair.

Continuous investigations by researchers have enriched our comprehension of traumatic brain injury, culminating in the development of a streamlined mechanism. They have utilized PP to manufacture flexible piezoelectric sheets to examine brain vibrations.¹¹¹ The FENG was coated with liquid electrical tape, as shown in Figure 7G. Subsequent activity from the meningeal body was recorded, as depicted in Figure 7H. Using particle image velocimetry and flexible sensors, information regarding internal strain and stress within the brain was collected, as illustrated in Figure 7I. An examination was conducted on the relationship between internal strain and stress information at two different time intervals, thereby validating the viscoelastic properties of the biological neural brain.

4 | SUMMARY AND FUTURE PROSPECTS

This paper provides a systematic review of the substantial advancements made by FENG in the realm of medical applications. As an emerging technology with immense potential, FENG, characterized by its high voltage coefficient, exceptional flexibility, and superior frequency response features, has showcased extensive applicability in the domain of implantable biosensing. This spans

FIGURE 6 Applications of ferroelectret nanogenerators in muscle movement. (A) Depicts basketball players employing intelligent kinesiology tapes to gather movement signals. In the future, this technology could allow coaches to wirelessly monitor and instruct their players. Reproduced with permission.⁷³ Copyright 2024, Elsevier. (B) Presents a flexible PP-piezoelectret film prepared for utilization, and (C) shows the recommended sensor placement locations. Reproduced with permission.⁵³ Copyright 2018, IEEE. (D) A PP-based tactile sensor on a wearable textile substrate, including a schematic demonstrating the ferroelectret tactile sensor system designed for gesture recognition and an image showing a ferroelectret tactile sensor used for finger bending. Reproduced under terms of the CC-BY license.¹¹⁰ Copyright 2022, published by MDPI. (E) The schematic diagram portrays the piezoelectret actuator/sensor structures. This wearable, flexible device could serve as a mechanical human-machine interface in applications such as real-time feedback and long-distance haptic communication. Reproduced with permission.¹²⁴ Copyright 2019, American Chemical Society. (F) Demonstration of the PTFLF-based sensors as wearable devices for self-powered body movement tracking, including the electric responses of the PTFLF-based sensor to a contact-release movement of a gripping glass beaker is shown and the electric responses of the PTFLF-based sensor to swallowing water are highlighted. Reproduced with permission.⁹⁶ Copyright 2022, Elsevier.

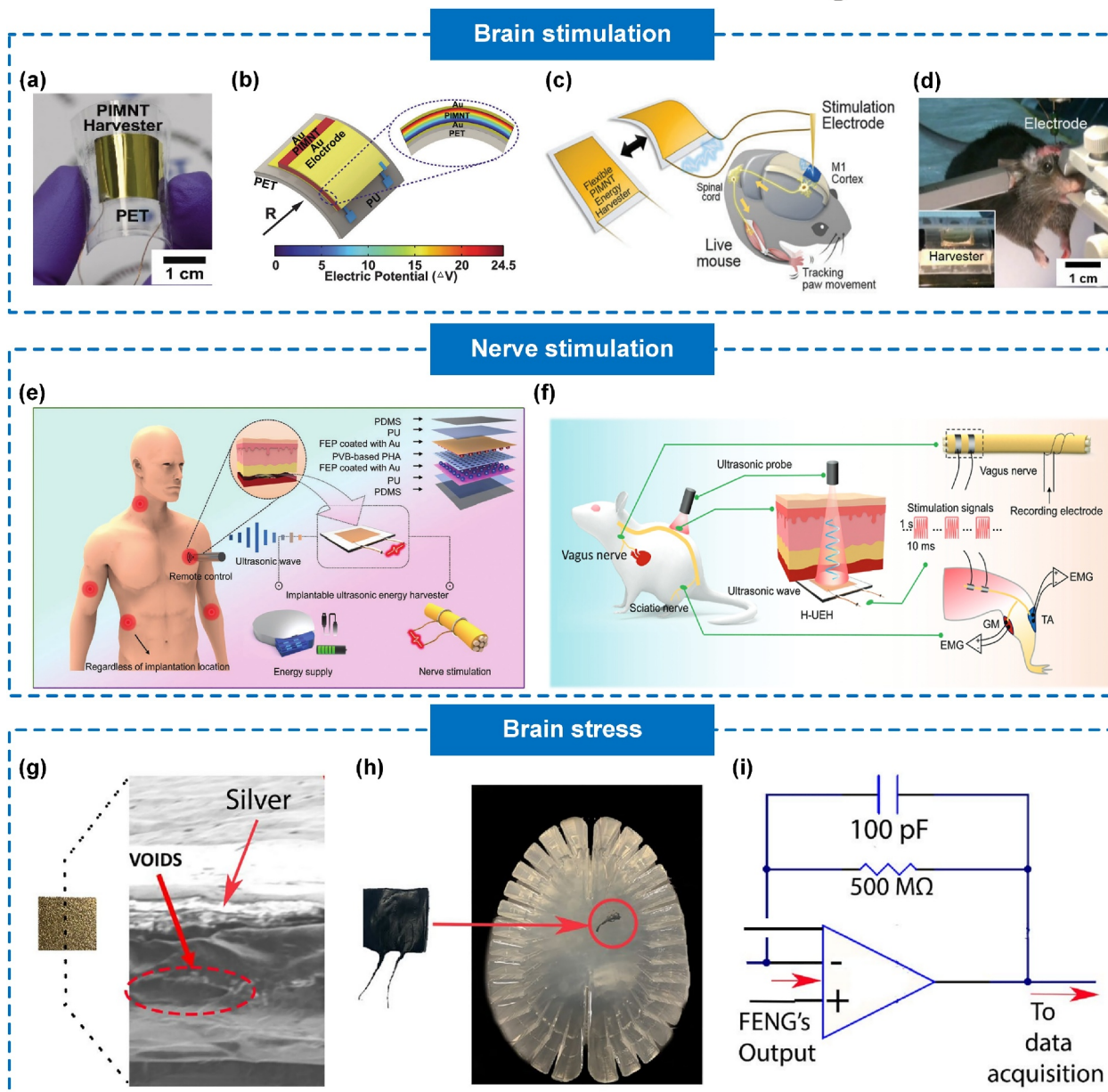


FIGURE 7 Applications of FENG in the neural reflex. (A) Photograph of the final flexible PIMNT harvesting device that is flexible enough to be completely bent by human fingers, (B) the theoretical simulation model of an MIM-type PIMNT thin film energy harvester, (C) the representation of an animal experiment where the flexible PIMNT harvester is used for brain stimulation in a mouse, and (D) the photograph of the PIMNT stimulator applied to the head of a mouse that has been immobilized and anesthetized with ketamine. Reproduced with permission.¹¹⁵ Copyright 2015, RSC. (E) Schematic of H-UEH as implantable bioelectronics for energy supply and neuroprosthetics and structure diagram of H-UEH, and (F) the schematic of the H-UEH driven by ultrasound for stimulating peripheral nerves. Reproduced with permission.¹¹⁶ Copyright 2022, Wiley-VCH Verlag. (G) FENG cross-section showing the voids and silver electrode, (H) FENG placed inside the brain phantom and (I) the charge mode amplifier circuitry to which the FENG output is fed. Reproduced under terms of the CC-BY license.¹¹¹ Copyright 2023, published by Springer Nature. FENG, ferroelectret nanogenerators.

across crucial medical areas including but not limited to blood circulation monitoring, respiratory system evaluation, muscle movement tracking, and neural reflex analysis, as depicted in Table 3.

While FENG technology holds the potential to significantly augment physiological efficiency, foster the

profound application of bionic electronic technology in numerous medical branches, and induce transformative changes in the medical health field, it concurrently encounters a series of challenges in the actual deployment process. These challenges primarily encompass aspects of biocompatibility, long-term stability, and safety, which

necessitate further comprehensive research and resolution. Moreover, the performance of materials utilized in FENG is susceptible to environmental elements such as temperature and humidity, which could result in inconsistent output performance, thereby impacting the overall efficiency and lifespan of the generator. When FENG is implanted into a biological body or positioned in a complex environment, biological buffering effects, electromagnetic interference, and potential immune rejection reactions could all negatively impact its performance. Despite FENG exhibiting numerous potential application prospects, its commercialization process remains relatively sluggish, principally constrained by technological maturity.

Moving forward, it is imperative for researchers to consistently delve into the discovery of new materials and design strategies aimed at improving the performance and safety of FENG technology. There is also a need to bolster clinical trials to affirm the long-term safety and efficacy of FENG within the human body. The relentless progression of material science offers a sturdy foundation for FENG's development. By fine-tuning the composition and structure of materials, enhancements can be made to its piezoelectric performance, biocompatibility, and long-term stability. For example, the investigation of new ferroelectric polar materials, which potentially exhibit higher piezoelectric coefficients, lower dielectric losses, and superior mechanical properties, could better fulfill medical application prerequisites such as high precision, sensitivity, and reliability. Simultaneously, the application of surface modification and coating technologies could augment the interfacial bonding force between FENG and biological tissues, mitigating the risk of immune rejection and enhancing its stability and safety within the body.

Advancements in biomedical research are poised to unveil new opportunities for FENG's medical applications. By gaining a comprehensive understanding of the physiological mechanisms of the nervous, muscular, and circulatory systems, researchers can more accurately design FENG's application scenarios and modes. For instance, within the realm of neuro-electric stimulation, FENG could serve as a novel self-powered stimulation source, used for the precise modulation of neuronal excitation and suppression states, thus offering fresh insights and approaches for diagnosing and treating nervous system disorders. Regarding muscle movement monitoring and assistive training, FENG can track the state and intensity of muscle activity in real-time, facilitating personalized training guidance and feedback for athletes and fitness enthusiasts.

Exploiting its non-ferromagnetic properties, miniaturization capabilities, and superior energy efficiency,

FENG shows vast potential in various sectors, including medical implantable devices, wearable sensors, and remote medical monitoring. Specifically, in the context of remote medical monitoring, FENG can function as a self-powered sensor that operates independently of external power sources, enabling continuous patient physiological parameter monitoring. It can transmit data to the cloud for analysis, providing medical professionals with immediate and precise diagnostic foundations and treatment recommendations. In addition, with the ongoing advancements in nanotechnology and micro-electronics, the integration and intelligence level of FENG is set to improve further. By pairing FENG with microprocessors and wireless communication modules, a smart medical system capable of autonomous perception, analysis, and decision-making can be developed, offering patients more comprehensive, personalized medical services.

In conclusion, FENG leverages the non-ferromagnetic, miniaturized, and energy-efficient properties of nanomaterials, offering distinct advantages such as lightweight design, compact size, low volume, and minimal power consumption. By directly converting energy into electricity, it demonstrates a faster response time and higher efficiency compared to conventional methods. Looking ahead, the future development of FENG is anticipated to achieve significant advancements and breakthroughs in various domains. Notably, FENG shows great potential for transformative applications, particularly in the field of medicine. By addressing current challenges and fully harnessing its potential, FENG could conceivably emerge as a novel and efficient diagnostic and therapeutic tool and technology platform, thus making significant contributions to human health.

AUTHOR CONTRIBUTIONS

Xianfa Cai: Conceptualization; funding acquisition; investigation; writing—original draft. **Xiaowen Han:** Data curation; investigation; writing—original draft. **Jiaqi Xie:** Conceptualization; funding acquisition; supervision; writing—review and editing. **Yunqi Cao:** Methodology; writing—review and editing. **Wei Li:** Conceptualization; project administration; supervision; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data will be available upon request from author.

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